



Driving Force and Pressure on the Food System of Remote Islands

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Abstract: Global development dynamics have a profound influence on food systems, particularly in remote island regions that are highly vulnerable to global market fluctuations and supply chain disruptions due to limited accessibility and dependence on external resources. This study aims to analyze the driving forces and pressures that shape the food system of remote islands. The analysis employs the Drivers Pressures State Impact Responses (DPSIR) framework, focusing on the components of driving forces and pressures. The findings reveal that the driving factors affecting island food systems are shaped by complex interactions among demographic, socio-cultural, economic, political, and biophysical dimensions. Meanwhile, environmental pressures influencing food availability are determined by three key aspects: agricultural systems, food resources, and retail structures. Pressures on local resources arise from unsustainable practices, including the burning of agricultural waste, uncontrolled livestock grazing, and destructive fishing methods such as fish bombing. Furthermore, the heavy dependence on food supplies from outside the island exacerbates the vulnerability of local food systems to logistical disruptions and the impacts of climate change. These findings underscore the urgent need for a fundamental transformation in agricultural practices to ensure the fulfillment of staple food needs while minimizing environmental pressures and enhancing the sustainability of island food systems.

Keywords: Driving force; Food system; Political; Pressures; Remote island

1 Introduction

Global development has a significant impact on the food system around the world [1]. Small Island Developing States (SIDS) are highly vulnerable to environmental pressures, particularly the impacts of climate change [2]. SIDS also face unique challenges due to their distinct characteristics, namely their small size and geographically isolated location, which result in limited economies of scale, reduced land area, high transportation costs, and significant food insecurity [2, 3]. SIDS are generally remote regions where the supply of protein sources relies heavily on fisheries. However, the fulfillment of staple food needs, particularly carbohydrates, remains suboptimal due to limited agricultural land and low local production capacity, resulting in a high dependence on food supplies from outside the islands [4]. This condition has long-term and sustained impacts on the well-being of communities living in remote islands. The SIDS in the Pacific region remain among the areas most dependent on external assistance [5]. This dependence not only reflects the limited capacity of local production and resources but also highlights the high degree of economic and social vulnerability. On the other hand, dependence on imported food can also make remote islands more vulnerable to global price fluctuation and supply disruption [6]. Availability of food in remote islands is influenced by a number of driving factors and pressures that are unique and complex. The driving factor on the food system of remote islands is divided into five main factors: (1) Environmental and biophysical driving factors. This unique biophysical condition is the key factor in determining the type of agriculture and fishery that can be developed on the island to achieve sustainable food security [7]. By understanding the local ecosystem and exploiting genetic diversity, varieties of plants and animals that are more adaptive and resistant to disease and climate change can be created, increasing long-term food security [8]. In addition, the development of environmentally friendly technology and innovation in resource management can be a solution to ensure that biophysics becomes a positive driver in achieving food security in the island region [9]. (2) Socio-cultural driving factors. The socio-culture of the

community in remote islands plays a role in shaping the food consumption pattern, and it also reflects the identity of a community [10, 11]. The reinforcement of the collaboration between farmers and fishermen can strengthen socio-cultural ties that support the local food security [12]. The integration of the socio-cultural aspect is the key in designing a holistic and sustainable food security policy in the island region [13]. (3) Demographic driving factor. The demographic driving factor includes isolated geographical conditions, limited access to resources, and challenges in transport and distribution [14]. This situation can cause the community in remote islands to be dependent, so that they become very vulnerable to global price fluctuation and supply disruption [15]. Therefore, an efficient distribution and logistics strategy must be adopted to ensure that the need for food can be met all around the island region, as well as to increase accessibility for the community in remote areas [16]. (4) Political driving factor. As the growth in population increases and the demand for food decreases, a policy that can accommodate this must be designed. Policies such as family planning programs and sex education can help manage population growth [17]. In addition, a holistic and forward-oriented policy is also key in creating an environment that supports food security in the islands [18]. Community-based agricultural support and the development of local institutions can be a part of the policy that encourages the active participation of the community in the management of food resources [19]. (5) Economic driving factors. Economic factors are a key element in designing a policy that supports food security in the islands [20]. Improvement in the service sector, sustainable tourism, or the creative industry can provide an economic alternative that can support food sustainability and security in the islands [21].

Pressure on the food system of remote islands is divided into three main factors: (1) the agricultural system. Through plant diversification and an agroecology approach, the people in the islands can create a more robust and environmentally friendly agricultural system, as well as improve adaptability to climate change [22]. Thus, building a sustainable agricultural system that is in accordance with the local environment is key to achieving food security in the islands without sacrificing environmental conservation [23]. By understanding and dealing with pressure factors on the environment resulting from food sources, sustainable food security in the islands can be created [24]. (2) Food source. The geographical condition that is isolated and distant from the center of food production limits the choice of available food, leading to a consumption pattern that tends to depend on certain food products [25]. Adopting a sustainable agricultural method, such as policulture, organic processing, and prudent land management, can be an important step in mitigating the pressure on the environment [26]. Therefore, it is important to apply a sustainable management policy [27]. (3) Retail. Improving logistics efficiency and adopting green transport technology can help to mitigate the environmental impact of the retail sector on the islands [28]. In addition, the initiative to recycle and limit the use of single-use plastic can help to mitigate the environmental impact from retail activity in the islands [29]. Thus, steps to mitigate the environmental impact of the retail sector are essential to achieve sustainable food security in the islands [30]. Based on this background, it can be seen that the food system in remote islands is greatly affected by the driving factors and pressure. Therefore, this research aims to analyze the driving factors and pressures that affect the food system of remote islands.

2 Method

The location of the research is in Mapila Village and Wumbulasa Village, which are located in the North Kabaena Subdistrict, Regency of Bombana, Province of Southeast Sulawesi, Indonesia. Both of these villages are located in remote islands that are far from the center of the city, hence they face a big challenge in accessibility and resource procurement, making it an area that is more vulnerable to a variety of external factors. The population of this research is the community, in this case, the head of family of a family who is responsible for the daily needs and lives in Mapila Village or Wumbulasa Village, Kabaena Island. The number of samples is determined by using the Krejcie and Morgan method with a 95% confidence level and a 5% margin of error [31]. Based on this method, the required sample size for a population of 585 households is 234 respondents. To minimize potential sampling errors, an additional 10% (equivalent to 23 respondents) was added, resulting in a total of 257 respondents, selected using a simple random sampling technique (126 Heads of Family in Mapila Village and 131 Heads of Family in Wumbulasa Village).

The research method that is used is the Participatory Rural Appraisal (PRA) and Drivers, Pressures, State, Impact, Responses (DPSIR) method. The PRA method is used to distribute questionnaires to the community, whereas the DPSIR method is an effective tool to formulate a more appropriate and sustainable policy in facing the challenges of sustainable development [32]. The DPSIR method has become a very significant paradigm in environmental studies and sustainable systems. This concept offers a comprehensive framework that allows an in-depth analysis of the complexity of the interaction between human activities and their impact on the environment [33, 34]. This is the beginning of an in-depth understanding of the dynamics that drive environmental change [35].

In this research, the DPSIR method is limited to the elements of the driving factors and pressure, relying on the DPSIR analysis framework approach that is explained qualitatively. The driving factors are human activities that trigger a variety of activities that can put pressure on the environment. The pressure arises as a result of a production or consumption process that is triggered by driving factors, namely, human actions taken in order to meet their needs.

The DPSIR model in the food system was proposed by the High Level Panel of Experts (HLPE) in 2017. This framework sought to explain how various factors and processes in the food system interact and contribute to the global food system, and it is related to the purpose of sustainable development (SDG). Furthermore, in this research, the global food system model will be adapted to the condition of the food security of the people in remote islands by analyzing a variety of driving factors, pressure, conditions, impact, and their response, which can be seen in Figure 1.

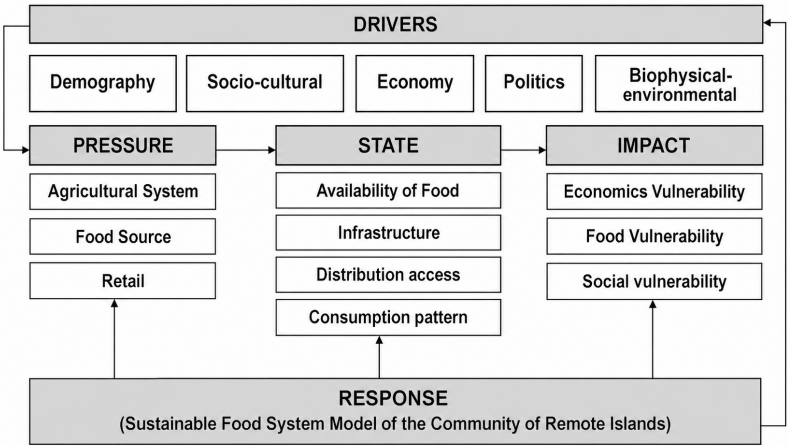


Figure 1. Framework food system, adapted from Ref. [36]

3 Results

3.1 The Findings of the Driving Force on the Food System in Remote Islands

Theoretical identification covers five factors that conceptually have the potential to be drivers, namely: demography, socio-cultural, economy, politics, and biophysical environment, that is described as follows:

(1) Demography

The demographic factor comprises population structure and population growth. The population structure in both Villages is a Constructive Pyramid in the form of an upside-down jar or a wasp's nest, which can be seen in Figure 2.

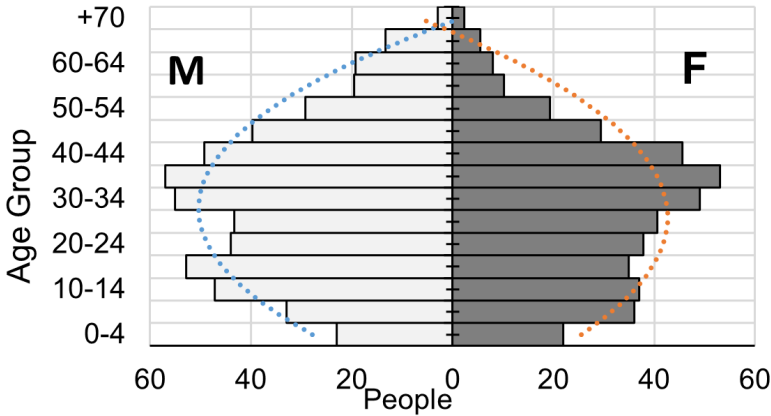


Figure 2. Pyramid of the population of Mapila and Mumbulasa Village

The population growth in both villages is calculated geometrically, showing a 1.45% population growth per year for Mapila Village, and a 4.15% growth per year in Wumbulusa Village. The high population growth in Wumbulusa Village can potentially increase the pressure on local resources, including the need for food and public services.

The characteristic of a constructive population pyramid is: the number of working-age population is relatively bigger than the number of children and parents, and the dependency ratio is low [37]. Based on this Constructive Pyramid, the value of the Dependency ratio in both villages can be known, namely 33.07%. This dependency ratio means that for every 100 persons in the working age group (usually defined as age 15–64), there are approximately 33 dependent persons (children under the working age as well as the elderly above the working age, usually defined as above 65). This dependency ratio is considered low, showing that the number of dependent people is smaller per working/productive age person, to potentially reduces the economic burden on the worker.

As shown in Figure 3, the population in both villages has continued to increase. This population growth is largely driven by migration, primarily stimulated by the expansion of the mining sector, which has created numerous employment opportunities in the surrounding areas. This condition has led to a growing demand for local food, particularly rice, which serves as the staple food for migrant communities. However, the demand for rice as the main staple cannot be met due to the limited availability of agricultural land suitable for paddy cultivation, resulting in a high dependence on food supplies from outside the island.

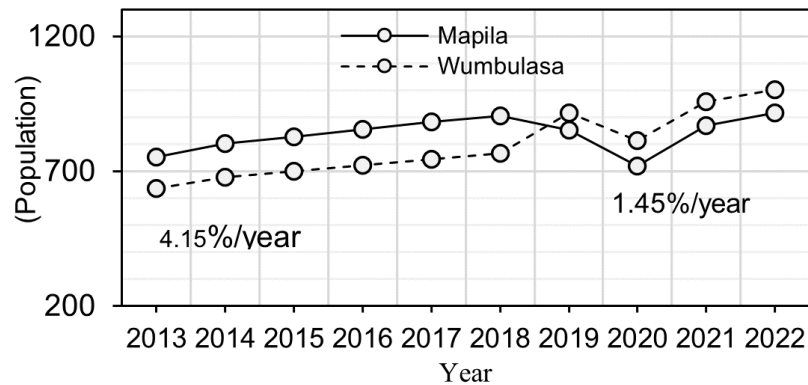


Figure 3. Population growth in Mapila Village and Wumbulasa Village

The population growth in Mapila Village and Wumbulasa Village reflects the demographic dynamics that can be used to encourage local development. This opportunity is very significant to increase economic productivity in both villages. This is in line with the research conducted by Lee and Handoyo et al., which suggests that the low dependency ratio found in both villages is a significant opportunity to increase economic productivity due to the more dominant working age population [38, 39]. However, if it is not followed by efforts to increase the workforce capacity and adequate work opportunities, this will become a challenge in the long term. There is a difference in the rate of population growth between Mapila Village and Wumbulasa Village, which is caused by several social, economic, and environmental factors that affect this dynamics. The growth is more rapid in Wumbulasa Village (4.15% per year), which can potentially increase pressure on local resources, including the need for food and public services, as explained by Huang et al. [40].

Therefore, the rapid population growth in rural areas entails an adaptive approach, such as efficient resource management and a community-based technology development to maintain the stability of food security [41]. The implication for the structure and dynamics of the population also indicates that there is a need for a strategic policy that supports the optimization of the local workforce. Skills training, institutional reinforcement such as provincially/municipally-owned corporation (Badan Usaha Milik Daerah, BUMDs), and the development of an appropriate technology can be the solution to exploit existing demographic potential. Therefore, there needs to be an integration approach between economic development and food security. This is in accordance with Naylor's research [42], which suggests that an integration approach between economic development and food security is extremely important, especially in regions characterized by remote islands. Based on this statement, it is important to manage sustainable population growth that should be accompanied by the development of the capacity of the local community to ensure that an inclusive and sustainable food system is achieved.

(2) Socio-cultural

Social and cultural practices that have taken root in the Kabaena community can affect the consumption patterns and food production. A more in-depth evaluation of this social aspect can be carried out by observing several forms of local wisdom in the family, community, and individual. Identification of local wisdom and individual lifestyle, household, and community is carried out based on Activities, Intensity, and Opinion (AIO). This evaluation measures opinions and the intensity of activities related to food consumption using a scale ranging from 0 to 100, where 0 indicates "strongly disagree" or "never," and 100 indicates "strongly agree" or "very frequently." This quantitative approach allows researchers to obtain a more objective overview of variations in consumption behavior among respondents, while also identifying general trends and differences between individual and household behaviors. The AIO scores for families and individuals under several lifestyle conditions are presented in Table 1.

Table 1 presents the overall results of the evaluation, illustrating the family's perspective on several aspects: food variety (consuming meals with a diverse range of dishes), rice always available (rice must always be present as a food ingredient), shared mealtime (family members must eat together), and rice as the main staple (rice is regarded as the primary staple food).

Table 1. Activity, Intensity, and Opinion (AIO) score of families and individuals on some lifestyle conditions in connection with food security [43]

Activity	Opinion (0–100)	Intensity (0–100)
Family		
Food variety	70.66	67.14
Rice always available	68.33	70.23
Shared mealtime	80.62	80.00
Rice as main staple	90.82	88.89
Individual		
Food variety	35.02	38.93
Shared mealtime	85.53	81.22
Rice as main staple	80.86	92.67

Note: opinion: strongly disagree (0)–strongly agree (100); intensity: never (0)–very frequently (100).

The difference in perception between family and individual, especially in food variety, while rice remains the main staple that is consumed with high frequency by both family as well as individual. The finding that is related to family and individual behaviour shows a high dependency on rice as the main staple, although the availability of food, particularly rice, is very limited in the island. This situation indicates a challenge in the sustainability of the local food system, where dependency on rice as the main staple is not matched by an adequate local production. Consequently, the people might have to depend on supplies from outside the island, which can potentially increase the vulnerability of food if distribution is disrupted. This is evident in the research conducted by Pane et al. [44], which suggests that the pattern of rice consumption at the household level is frequently influenced by socio-cultural factors, including hereditary habits and values inherent in rice consumption as a symbol of social status.

In addition, low food variety that is visible at the individual level also confirms the limitation in food diversification, which can aggravate the risk of instability if the supply of rice is disrupted. The preference for rice and the low variety of food is a reflection of a weak consumption pattern, which is vulnerable to a limited food supply in the island, which has an effect on the risk of malnutrition owing to a lack of food variety. This is in line with the research conducted by Gunadi et al. [45], which suggests that preference for certain main staples, such as rice, often hampers the people's acceptance of local food alternatives that are more nutritious. In addition, the research conducted by Sudrajat [46] also reveals that the diversification of food consumption can increase the community's food security by mitigating the vulnerability to the disruption of staple food supply, especially in remote areas. In addition, certain cultural patterns related to food consumption (such as a priority to processed food and tobacco) can aggravate nutritional imbalance in the family. This is in line with the research conducted by Watts et al. [47], that suggests that lack of education can have an impact on imbalanced food pattern, and the research conducted by Khasanah and Sumarmi [48] that indicated that the existence of food diversification constraint in numerous rural areas in Indonesia is affected by the level of education, income, and access to information. A consumption culture that prioritizes processed food also increases the difficulties in food security. As stated in the research by Lestari and Saidah [49], the habit of consuming processed food is often driven by the modernization of living patterns, but this can cause nutritional imbalance in the people in rural areas. Therefore, an intervention that emphasizes nutrition education and the promotion of local food diversification is very necessary to encourage a balanced and sustainable food consumption pattern in remote islands [50, 51], including nutrition education that provides an understanding for the community about the importance of a balanced nutrition [52], and promoting a sustainable farming practice and plant diversification to enrich the choice of local food [53]. The success of this diversification depends so much on the ability of the community to access technology and infrastructure that supports the production and distribution of local food [54]. This step is also supported by the research conducted by Sutrisno et al. [55], which indicates the importance of a socio-cultural approach in increasing food security, especially through the utilization of the local food potential as an alternative.

The tradition in managing land, the traditional agricultural system, ways to raise livestock, and fishing in certain ways are the parts that are evaluated by the social aspect drivers. Some description of traditions that are based on the results of the interview and identified PR through content analysis that is considered to have a negative impact on the environment, or means that are unable to produce good food plants, is presented in Table 2.

(3) Economic

Economic conditions—such as household income levels, availability of employment opportunities, and dependence on specific sectors—significantly influence food access and availability. Low income levels can limit a household's ability to purchase nutritious food, particularly when a large proportion of income is allocated to non-food expenditures. The lack of economic diversification and limited employment opportunities further exacerbate this issue. Moreover, an economy that relies heavily on food supplies from outside the island increases the vulnerability of the local food system to price fluctuations and supply instability. The distribution of respondents by monthly average spending group for spending on food and non-food can be viewed in Table 3.

Table 2. The tradition of fishing, plant cultivation, and livestock is considered negatively, but is still practiced

Food Production Activity	Description of the Tradition of the Local Community
Going to sea/fishing	Some members of the community (fishermen) still use bombs or poison when fishing. The use of fish bombs is considered effective and efficient because it results in a big catch at one stroke; explosives are “not difficult to obtain” and “easy to assemble”, although the risk has materialized (accident), it was considered that this accident was due to negligence on the part of the fisherman.
Corn farming/horticulture	Agricultural waste, such as corn straw, is not directly utilized or processed further for livestock needs or making organic fertilizer. Straw is commonly left scattered on the field or burned. This is not considered a problem because it will later be eaten by livestock or weathered into the earth. However, the burning of waste is often carried out because it can improve soil fertility.
Cashew farming	Cashew fields are usually left unattended, the edge of the field is overgrown with bushes (such as Jonga-jongan (<i>Chromolaena odorata</i>), and Lantana Camara (marica-marica in local term, <i>Lantana camara</i>). The bushes will be removed as harvest approaches because these bushes will hinder the harvest process. The removal of bushes is usually carried out by means of chopping; only a small part will be burned.
Livestock farming	All types of livestock are kept by releasing them to roam freely. This practice has become a tradition, although it has lately reaped problems, because livestock that are released often enter fields. However, because there are only a small number of farmers who plant crops, while the cultivated land is narrow (can be fenced), this is a problem that can be solved easily. As for livestock droppings that are scattered along the road (especially cow droppings), it is admittedly unfortunate, but so far, no action has been taken.

Table 3. Distribution of respondents by monthly average spending group for spending on food and non-food

Monthly Spending Group	Respondent Percentage by Village		Total (%)
	Mapila (%)	Wumbulasa (%)	
Foodstuff			
<IDR 400,000	3.97	3.05	3.50
IDR 400,000–1,000,000	18.25	12.21	15.18
IDR 1,000,000–1,500,000	27.78	22.90	25.29
IDR 1,500,000–2,000,000	15.87	31.30	23.74
>IDR 2,000,000	34.13	30.53	32.30
Total	100.00	100.00	100.00
Non-food item			
<IDR 400,000	31.75	35.11	33.46
IDR 400,000–1,000,000	30.16	25.19	27.63
IDR 1,000,000–1,500,000	12.70	12.98	12.84
IDR 1,500,000–2,000,000	4.76	13.74	9.34
>IDR 2,000,000	20.63	12.98	16.73
Total	100.00	100.00	100.00

Based on Table 3, if it is linked to the poverty standard that refers to a spending of less than IDR 400,000 per month (the average number of the poverty line for the Regency of Bombana for foodstuffs), which shows that most people allocate a budget of more than IDR 400,000 per month for non-food needs, which means that they are above the spending limit, which is one of the indicators of poverty. Larger spending, especially in a range of IDR 1,000,000 to more than IDR 2,000,000 for foodstuff indicates a purchasing power that is quite strong. Nevertheless, the high level of expenditure also indicates that access to food sources remains inadequate, resulting in the majority of household income being absorbed by food-related needs. Based on this observation, it can be inferred that the communities in both villages generally do not experience economic conditions indicative of extreme poverty, particularly in terms of fulfilling their basic food requirements.

This finding is also consistent with the Food and Agriculture Organization’s (FAO) Engel Coefficient theory. In terms of proportion, household expenditure on food accounts for approximately 83% more than expenditure on non-food needs. According to the FAO, this can be analyzed using the Engel Coefficient, which represents the proportion of food expenditure to total household expenditure. When the Engel Coefficient exceeds 59%, a household

is categorized as being in poverty or highly vulnerable, as the majority of its income is spent solely on meeting basic food needs, leaving limited capacity for other essential expenses such as education, healthcare, or savings.

This is in line with the research by French et al. [56] and Atzori et al. [57] that suggests that the income elasticity for the demand for food in developing countries is often related to an increase in the consumption of quality and varied food, and the importance of the food spending pattern as an indicator of household prosperity, where a higher allocation for foodstuff usually reflects a level of food security that is higher.

With regard to the pattern of foodstuff and non-food spending in Mapila Village and Wumbulasa Village, it has been found that dependency on foodstuff, especially rice, is very high in both villages. This indicates a potential vulnerability to price or food supply. Meanwhile, lower non-food spending probably reflects limited access to goods and non-food services, which must also be taken into consideration in developing the local economy. This is in line with the research by Hafizah et al. [58] and Langford [59], which indicates that food security is affected not only by spending, but also by distribution and food availability in certain areas. Therefore, there needs to be a policy strategy that integrates the improvement of infrastructure, technology, and the diversification of local food can be an important step in ensuring a sustainable food security in remote islands.

(4) Politics

The evaluation of the driving factors on political aspects is measured based on the community's perception of aid/support that they have received from the government and private parties. Support from the private sector in this context primarily includes Corporate Social Responsibility (CSR) programs implemented by companies, which play a role in promoting community welfare and strengthening food security in these remote island regions. The results of the assessment regarding public perception of these forms of support as shown in Table 4.

Table 4. Respondents' perception score on several scopes of governmental and private support/aid in food security

Scope of Policy and Cooperation	Village		Total (%)
	Mapila (%)	Wumbulasa (%)	
Government support/aid			
Increase production of crop	65.6	74.0	69.9
Increase production of plantation crop	64.9	72.8	68.9
Increase production of livestock	64.9	72.5	68.8
Availability of main staple	64.9	62.1	63.5
Average			67.78
Private support/aid			
Increase production of crop	82.4	78.9	80.6
Increase production of plantation crop	80.3	79.1	79.7
Increase production of livestock	77.5	74.2	75.8
Availability of main staple	63.7	59.5	61.6
Average			74.43
Partnership/cooperation			
Willingness of food production partner	86.0	85.5	85.8
Preference for cash aid	68.1	60.2	64.0

Based on Table 4, the findings indicate that the community perceives the role of private support or aid as more dominant compared to government support. This is reflected in the average scores given by the respondents, where private sector initiatives received a higher rating of 74.43% compared to those of the government. The perception score that results from the survey of the people in Mapila Village and Wumbulasa Village shows a great appreciation for the support, but it also suggests the need for improvement, especially in the availability of the main staple and other food. The implication of this finding shows that by reinforcing the collaboration among the government, private sector, dan the community, food security in both villages can be improved further. This is in line with the research by Wang et al. [60] that suggests that good governance, including subsidized distribution and infrastructure development, is very important to ensure food accessibility and availability. The lack of infrastructure is often the main obstacle in the distribution of food in remote areas, and the lack of government support in the distribution of the main staple [57], although the community's perception of government support is quite positive. Therefore, the public-private partnership is important to increase food availability, especially in areas with limited access [61]. An effective collaboration can maximize existing resources, particularly in supporting the production of local food, with adequate political stability, the distribution of food and management of commodity price can be more guaranteed [62, 63]. Reinforcement of the collaboration among parties and improvement of the quality of infrastructure are important steps to support food security in both villages.

(5) Biophysical environments

The objective evaluation of biophysical factors is obtained through the observation on the response of farmers and fishermen to the effect of climate change on production activity in both villages, which can be viewed in Table 5.

Table 5. Perception score of farmers and fishermen on the influence of the weather/climate on the activity of farming/fishing

Perception of Biophysical Condition	Village		Total (%)
	Mapila (%)	Wumbulasa (%)	
The weather/climate conditions on the island in the last five years influenced/decreased the production level of plants cultivated	73.9	83.2	80.3
The supply of water for agricultural land is adequate for production activity for one year	46.1	49.6	48.5
The weather/climate conditions of the island in the last five years influenced/decreased the amount of catch in the sea	78.7	78.2	78.5

The community's perception in Mapila Village and Wumbulasa Village on the condition of the biophysical environment shows that climate change is perceived as a significant pressure on both villages, with an average score of 80.3, therefore it affects the agricultural and fishery productivity in the last five years. Supply of water for farming is considered inadequate in both villages, with an average score of 48.5.

There is no access to irrigation for farming in both villages, while the river only flows in the southeastern part of the village. At the same time, evaluation of the development of rainfall shows a rise in dry months (rainfall < 100 mm/month) during 2022 and 2023. In 2022, the total rainfall was 2,253.4 mm, falling to 1,384 mm in 2023. This condition indicates that the limited supply of water is the main challenge that limits the productivity of plants that in turn brings real pressure on the agricultural and fishery sector in both villages in the island. This is in accordance with the result conducted by Hidayat and Imelda [64], that suggests that the instability of the biophysical environment condition can disrupt the agricultural cycle and decrease the productivity of plants. Therefore, there needs to be an adaptation strategy to mitigate the impact of climate change and increase the access and management of water resources.

In addition, the dependence on external food supplies, particularly rice, creates significant pressures, primarily in the form of increased carbon footprints resulting from food transportation processes. The greater the distance traveled, the higher the greenhouse gas emissions produced from the use of fossil fuels in both maritime and land transportation modes. This condition not only adds to the environmental burden but also reflects a high dependency on external supply sources, which ultimately undermines the sustainability of the local food system.

Furthermore, the weather and climate condition is considered to affect the decrease in the amount of catch in the sea with an average score of 78.5. This shows that there is a significant influence on fishery products, which is triggered by a change in climate and weather over the last years. This is in accordance with the research by N'Souvi et al. [65] and Ojea et al. [66], which suggests that climate change also affects the sea ecosystem, which has an impact on the drop in the total catch for the coastal community, particularly for fishermen, therefore it becomes hard for them to survive and maintain their economy. Therefore, there needs to be an adaptation strategy for the coastal community, such as diversification of the source of income and a change in the method of fishing.

Based on the five identified driving factors, it can be concluded that politics is the most influential factor in determining the sustainability of food systems in remote islands. Political policies that support the national food system play a crucial role not only in maintaining the balance between production and consumption but also in ensuring environmental sustainability and community well-being. Thus, although demographic, socio-cultural, economic, and biophysical environmental factors also contribute, political policy exerts the greatest influence, as it serves as the foundation that regulates and integrates all these aspects into a coherent and sustainable food system.

3.2 Pressure on the Food System in Remote Islands

Pressure on the environment is divided into three, namely the agricultural system, food source, and retail, which are described as follows:

(1) Agricultural system

The agricultural potential in this research is measured based on three main aspects, namely: 1) agricultural land; 2) human resources (farmers), and 3) types of commodities and the size of production. Data on the agricultural land area refers to an official source (secondary data) available in the region, namely the BPS-Regency Kendari publications 2020–2024. The data was later compiled with the independent data of the evaluation result based on land use classification by the SENTINEL 2 Satellite image (ubin N0511-R060). The indication of human resources

potential, referring to the result of the Agricultural Census of 2023 that has been released by BPS-Nasional and BPS-regency Bombana, the number of farmers is counted based on the indicator of Number of Individual Agricultural Business. BPS categorizes types of agricultural and marine businesses by sub-sectors and type of commodity. The agricultural sector includes: sub-sectors of crops, secondary crops, horticulture, plantation, and livestock raising. It is of note to underline that the indicator of the number of individual business units is not completely accurate in quantifying farmers' human resources, because it is the number of business units that is counted. Based on the assumption used by BPS-Bombana [67], "The Number of Individual Agricultural businesses equates with the number of individual business units that carry out farming activities that produce crop products (paddy and secondary crop), including the crop breeding business, and not as a farm labor or family worker". Based on this assumption, although a business unit is run by more than one person, it will be considered as one business unit based on the business owner/farmer. An individual businessman (farmer) may manage more than one type of agricultural business unit, for example, a farmer who owns an individual crop business may also run a livestock business, and/or a plantation. Based on this, the number of "farmers" is still compared with the result of the field questionnaire survey, and all information about agricultural potential on all aspects is complementary.

(2) Food source

The review scope of the agricultural and marine potential, in connection with food security, apparently needs to be expanded. Previously focused on the scope of North Kabaena Regency, it should now be expanded to all administrative districts of the Bombana Regency on the Kabaena Island. There are 6 subdistrict areas in the Kabaena Island, namely: the subdistrict of Kabaena, North Kabaena, South Kabaena, West Kabaena, East Kabaena, and Central Kabaena. It is hard to find information on detailed agricultural land in official publications, including from the BPS-Bombana Regency source, as well as the local government. Based on the BPS publication of 2024, from 2017 to 2024, the rice field area in the subdistrict of North Kabaena has not changed, staying put at 415 ha. The writer confirmed the data on the land area by comparing the evaluation of land cover (supervised classification) using the data from the SENTINEL-2 satellite image of 13 August 2024, as presented in Figure 4.

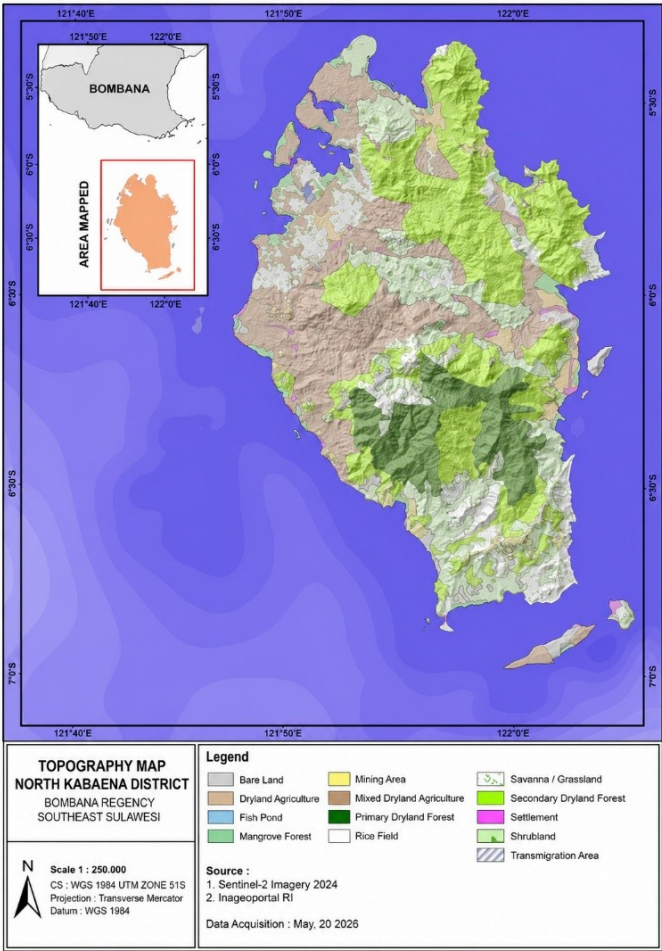


Figure 4. Map of land use in Kabaena Island

The evaluation of land use shows three forms of the most extensive land use in Kabaena Island, namely: secondary dryland forest (31.7%); mixed dryland farming (25.4%); and thicket (13.6%); while the the narrowest form of land use is rice field, and this land use can only be found in the subdistrict of North Kabaena. Specifically in the subdistrict of North Kabaena, the rice field area in 2024 is around 106.9 ha (0.9%); the use of dryland farming amounts to 480.6 ha (3.8%); and mixed dryland farming amounts to 3,858.4 ha (30.8%), which can be viewed in Table 6.

Table 6. Land use in Kabaena Island Regency of Bombana by Subdistrict area

Land Use	Subdistrict						Kabaena Island*
	Kb	KbU	KbS	KbTg	KbT	KbB	
	ha (%)						
Primary dryland forest	837.7 (8.1)	0.0 (0.0)	3,924.4 (32.4)	2,348.2 (7.8)	1,778.7 (18.8)	0.0 (0.0)	8,889.0 (11.1)
Secondary dryland forest	1,331.4 (12.8)	2,481.2 (19.8)	2,182.2 (18.0)	15,948.7 (52.9)	2,255.1 (23.9)	1,233.2 (21.6)	25,431.8 (31.7)
Primary/secondary mangrove forest	8.0 (0.1)	494.3 (3.9)	225.9 (1.9)	217.7 (0.7)	306.9 (3.2)	511.8 (9.0)	1,764.5 (2.2)
Thicket	1,068.5 (10.3)	2,878.6 (22.9)	1,037.5 (8.6)	3,961.2 (13.2)	1,829.3 (19.4)	146.3 (2.6)	10,921.4 (13.6)
Savana	1,011.2 (9.7)	2,089.9 (16.7)	274.9 (2.3)	2,415.2 (8.0)	504.9 (5.3)	1,193.0 (20.9)	7,489.1 (9.3)
Dryland farming	0.0 (0.0)	480.6 (3.8)	0.0 (0.0)	569.9 (1.9)	822.1 (8.7)	24.6 (0.4)	1,897.2 (2.4)
Mixed dryland farming	5,826.5 (56.0)	3,858.4 (30.8)	3,344.2 (27.6)	4,041.8 (13.4)	984.1 (10.4)	2,333.0 (40.8)	20,388.0 (25.4)
Rice field	0.0 (0.0)	106.9 (0.9)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	106.9 (0.1)
Pond	0.0 (0.0)	0.0 (0.0)	5.0 (0.0)	0.0 (0.0)	35.7 (0.4)	29.3 (0.5)	70.0 (0.1)
Settlement	36.0 (0.3)	31.0 (0.2)	61.5 (0.5)	100.7 (0.3)	138.3 (1.5)	66.3 (1.2)	433.8 (0.5)
Mining	95.6 (0.9)	0.0 (0.0)	529.1 (4.4)	204.8 (0.7)	77.0 (0.8)	6.5 (0.1)	913.0 (1.1)
Open land	187.7 (1.8)	122.8 (1.0)	514.4 (4.3)	314.1 (1.0)	712.7 (7.5)	169.8 (3.0)	2,021.5 (2.5)
Total	10,402.5 (100.0)	12,543.6 (100.0)	12,099.1 (100.0)	30,122.3 (100.0)	9,444.8 (100.0)	5,713.8 (100.0)	80,326.1 (100.0)

Note: Kb = Kabaena District; KbU = North Kabaena District; KbS = South Kabaena District; KbTg = Central Kabaena District; KbT = East Kabaena District; and KbB = West Kabaena District; *Total area of the districts included in the Bombana Regency area.

Table 6 illustrates the form of land use in Kabaena Island, Bombana Regency, by area in every subdistrict. Based on Table 6, the rice field area in Kabaena is recorded at 106.9 hectares. When compared to the island's total consumption needs, this area can only meet approximately 25% of the overall demand, indicating a significant deficit in local rice supply. This condition demonstrates that local production capacity has not been able to keep pace with the rapid population growth and the steadily increasing demand for staple food. Furthermore, the rice produced by local farmers is not entirely distributed to meet local market needs. A substantial portion is exported off the island due to its relatively high quality and greater market value. Consequently, high-quality local rice has become relatively scarce in local markets.

In this situation, the people of Kabaena Island tend to prefer rice imported from outside the island, which is cheaper, even though its quality is lower than that of locally produced rice. This phenomenon reflects a mismatch between agricultural production potential and local distribution and consumption patterns. On one hand, farmers are economically motivated to sell their products outside the region for higher profits, while on the other hand, local communities face limited access to quality food. If this condition persists, food security on Kabaena Island will become increasingly vulnerable to market fluctuations and dependence on external supplies.

Furthermore, information on agricultural human resources based on the Agricultural Census of 2023 in Kabaena Island Bombana Regency, by area in each subdistrict can be viewed in Table 7. Table 7 shows the number of individual agricultural businesses in the entire area of Kabaena Island, Bombana Regency, which amounts to 8,286

business units; most of these business units are in the plantation subsector (48%); then livestock (34%), horticulture (9.6%), crops (5.1%), and secondary crops (3%). The distribution of this individual business suggests that most farmers in Kabaena Island carry out production activity in the plantation sub-sector, and only a few of them produce crops/secondary crops.

Table 7. The number of individual agricultural businesses (farmers) in several subdistricts in Kabaena Island from the Agricultural Census 2023, by agricultural sub-sector

Subdistrict on the Island	Farmers by Sub-Sector (person)				
	Crop	Secondary Crops	Horticulture	Plantation	Farming
Kb	15	15	34	600	365
KbU	144	22	123	626	568
KbS	35	35	31	563	220
KbB	68	41	74	707	507
KbT	78	74	421	706	563
KbTg	80	65	116	780	610
Kabaena Island	420	252	799	3,982	2,833
Total	8,286 business units				

Note: Kb = Kabaena District; KbU = North Kabaena District; KbS = South Kabaena District; KbTg = Central Kabaena District; KbT = East Kabaena District; and KbB = West Kabaena District.

The activity of producing crops is still low, compared to the total of 8,286 agricultural businesses in Kabaena Island. This reflects the low dependency on food agriculture in Kabaena, with most farmers likely involved in other agricultural or non-agricultural sectors. This can be an indication of the need to improve the management of agricultural resources to increase productivity and food security in the region.

(3) Retail

The retail sector in Mapila and Wumbulasa Village plays a vital role in supporting the economy of the local community, although both of them are villages that are located in remote areas in the Bombana Regency. The economic life of the people in these villages is highly dependent on the retail sector that provides various daily necessities from foodstuffs to household items. The retail sector in both villages has developed quite well, thanks to the high demand for daily necessities. The traditional market around these two villages also functions as a trade center, where people can buy fresh fish, meat, etc, that is commonly supplied by farmers and fishermen from surrounding villages. Clothes, farming tools, and local products are also produced by the local people. Grocery stores, stalls, and small kiosks spread across both villages and are the main places for the people to obtain basic materials such as rice, oil, sugar, flour, and other household necessities. Some stalls sell other groceries such as vegetables, fruits, eggs, and meat, that is usually supplied directly by local farmers. Besides grocery stores, kiosks, and permanent stalls, some peddlers travel around from one village to the other. The presence of peddlers who sell a variety of items makes an important contribution to facilitating people's access to needed items. This allows people to access a variety of goods without having to travel far to big markets. These peddlers often bring goods from nearby towns or even from outside the region, goods that are not available in stalls or local stores, and sell them directly to consumers at a more affordable price and without high additional costs.

On the other hand, the retail sector in Mapila and Wumbulasa Village is also confronted with various challenges. One of the biggest challenges is limited infrastructure, especially transportation that connects the village to the distribution center of goods. This constraint in transportation causes the price of goods sold in the local market to be higher than the price in the market in big cities, because of the distribution cost that must be paid by traders. In addition, although modern markets and big retail stores are starting to appear in several areas of the Bombana Regency, people in Mapila and Wumbulasa Village prefer to shop in traditional markets and local stalls, which allow them to interact directly with the traders and buy goods at a more flexible price. This shows that although the influence of the modern market is becoming stronger, traditional markets and a trading system that is based on social and trust relations still have an important position in both villages.

The trading pattern that is interdependent among local traders, consumers, and producers in Mapila and Wumbulasa Village still runs smoothly, with the people supporting each other. A lot of village residents prefer to buy goods from stalls or traders with whom they are acquainted, and the transaction is often carried out with a credit system or payment in instalments, which allows people with limited income to meet their needs. Although the retail sector in the village is still dominated by small-scale trading and traditional ways, there is a tendency that the retail sector

will develop further as access to education and technology improves, which allows the people to be connected to a more extensive market. Overall, the retail sector in Mapila and Wumbulasa Village plays a vital role in the social and economic life of the village community, by maintaining traditional characteristics that have long existed and facing challenges that come along with current development.

The discussion of pressure on the food system in remote islands: Pressure on local resources is a result of unsustainable practices such as incineration of agricultural waste, the release of wild livestock, and the exploitation of waters with fish bombing. Great dependence on food supply from outside the island increases the vulnerability of the food system to logistical disruption and climate change. This unsustainable food system demands a fundamental transformation in agricultural and fishery practices to reduce pressure on the environment. The isolated islands can become independent of other islands for food if their full potential is maximised [68]. Therefore, in order to develop a resilient and sustainable food system, a comprehensive and context-specific approach is required. This transformation may be necessary in the form of greater localised expertise in the region [69].

4 Conclusions

The role of the driving factors in the food system of remote islands is influenced by a complex interaction between findings, namely: (1) demography: the dependency ratio in both villages is 33.07%, indicating a relatively low dependency level. This suggests that the number of dependents is smaller than the working-age population, which has the potential to reduce the economic burden on productive individuals; (2) socio-cultural: based on AIO, families and individuals depend highly on rice as the main staple, although the availability of food, especially rice, is very limited in the island. The limited food variety is a reflection of a weak consumption pattern, that is vulnerable to limited food supply in the island, and implies malnutrition due to lack of food variation; (3) economy: most of the people allocate their expenditure more than IDR 400,000 per month for non-food needs, while spending at a range of IDR 1,000,000 to more than IDR 2,000,000 that is allocated for foodstuff. The finding regarding the difference in foodstuff and non-food spending is due to the dependency on foodstuff, especially rice, which is very high in both villages; (4) politics: there is a great appreciation towards the support for improvement by reinforcing the collaboration between the government, private sector, and the community, therefore the food security in both villages can be further improved; (5) the environment: it shows that there is climate change, inadequate water supply for farming, and weather condition that is perceived as a significant pressure on both villages. Meanwhile, the role of pressures in meeting the need for food leads to some findings: (1) the agricultural system, that is measured by three main aspects, namely: agricultural land; human resources (farmers), and type of commodity and size of production; (2) food source, consists of three forms of land use, namely: secondary dryland forest, mixed dryland farming, and thickets; with most of these business units being in the sub-sector of plantation, livestock, horticulture, crops and secondary crops; (3) retail, such as traditional market, grocery stores, stalls, small kiosks, and peddlers that can support the economy of local people.

Author Contributions

Conceptualization, N.I.A., T.E.B.S., E.F., and S.W.U.; methodology, N.I.A., T.E.B.S., E.F., and S.W.U.; software, N.I.A.; validation, N.I.A., T.E.B.S., E.F., and S.W.U.; formal analysis, N.I.A., T.E.B.S., E.F., and S.W.U.; investigation, N.I.A.; resources, N.I.A.; data curation, N.I.A.; writing—original draft preparation, N.I.A., T.E.B.S., E.F., and S.W.U.; writing—review and editing, N.I.A., T.E.B.S., E.F., and S.W.U.; visualization, N.I.A.; supervision, T.E.B.S., E.F., and S.W.U.; project administration, N.I.A.; funding acquisition, N.I.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

AIO	Activities, Intensity, and Opinion
BPS	Statistics Indonesia Kaur Regency
BUMDs	Provincially/municipally-owned corporation
CSR	Corporate Social Responsibility
DPSI	Drivers, Pressures, State, Impact
DPSIR	Drivers, Pressures, State, Impact, Response
Kb	Kabaena District
KbB	West Kabaena District
KbS	South Kabaena District
KbT	East Kabaena District
KbTg	Central Kabaena District
KbU	North Kabaena District